

Property-preserving numerical methods for the shallow-water equations with source terms

Hennes Hajduk

Institute of Applied Mathematics, TU Dortmund University, Germany

MFEM COMMUNITY WORKSHOP

September 25, 2026

- 1 Introduction
- 2 Limiting for systems of conservation laws
- 3 Extension to shallow water equations with topography
- 4 Conclusions

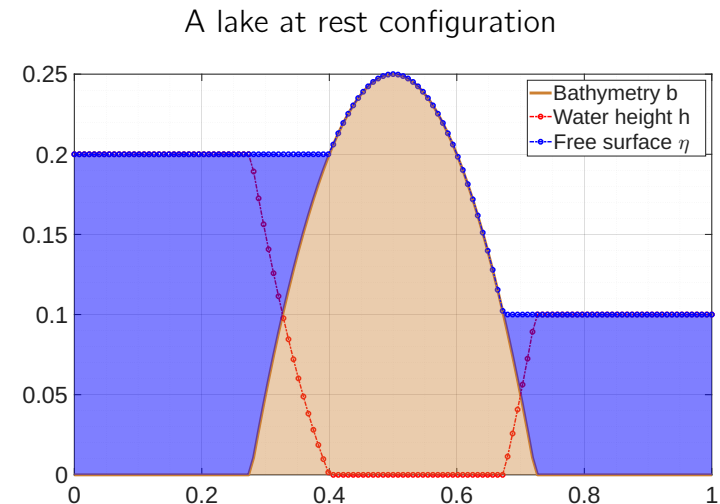
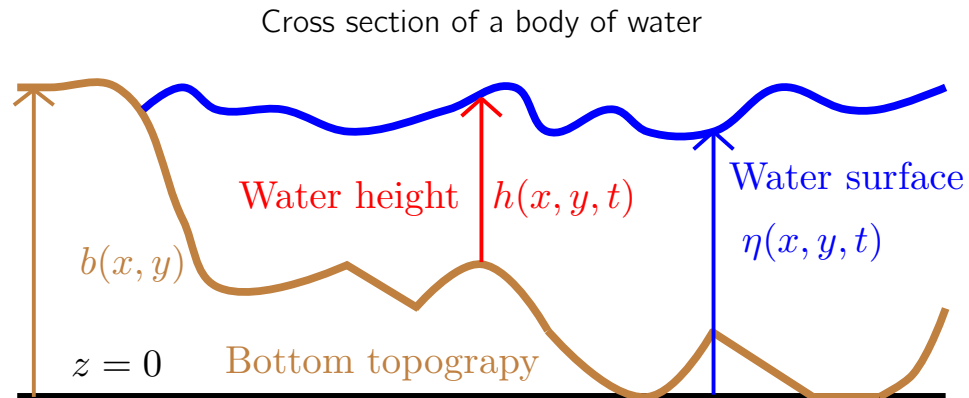
SHALLOW WATER EQUATIONS (SWE)

- Hyperbolic system of balance laws

$$\frac{\partial}{\partial t} \begin{bmatrix} h \\ h\mathbf{v} \end{bmatrix} + \nabla \cdot \begin{bmatrix} h\mathbf{v} \\ h\mathbf{v} \otimes \mathbf{v} + \frac{g}{2}h^2\mathbf{I} \end{bmatrix} + \begin{bmatrix} 0 \\ gh\nabla b \end{bmatrix} = 0,$$

- Unknowns: water height, depth-integrated velocity vector $\mathbf{v}$; parameters: gravity $g = 9.81 \frac{\text{m}}{\text{s}^2}$, bottom topography $b = b(\mathbf{x})$
- Well-balancing: preserve *lake at rest*

$$\mathbf{v} \equiv \mathbf{0}, \quad \frac{g}{2}\nabla h^2 + gh\nabla b = gh\nabla\eta \equiv \mathbf{0}$$



GOALS OF THIS WORK

- Develop **property-preserving** finite element methods for balance laws of the form

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{f}(u) + s(u, b) = 0$$

- Approximations u_h should preserve the following properties of exact solutions u

- ▶ Conservation

$$\int_K u(\mathbf{x}, T) \, d\mathbf{x} = \int_K u(\mathbf{x}, 0) \, d\mathbf{x} - \int_0^T \int_{\partial K} \mathbf{f}(u(\mathbf{x}, t)) \cdot \mathbf{n} \, d\mathbf{s} \, dt - \int_0^T \int_K s(u, b) \, d\mathbf{x} \, dt$$

- ▶ Physical/mathematical constraints, i. e., $h \geq 0$ for shallow water equations
- ▶ Lack of spurious oscillations within global bounds (no Gibbs phenomena)
- ▶ Discrete versions of entropy inequalities

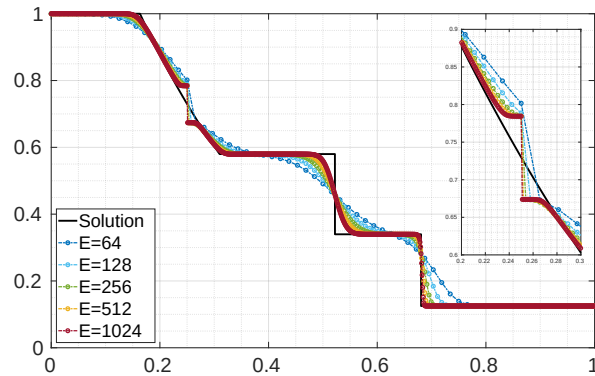
$$\frac{\partial U(u, b)}{\partial t} + \nabla \cdot \mathbf{F}(u, b) \leq 0, \quad \mathbf{F}_u = U_u^\top \mathbf{f}_u, \quad \mathbf{F}_b = U_u^\top s$$

- ▶ Preservation of equilibria (well-balancedness)

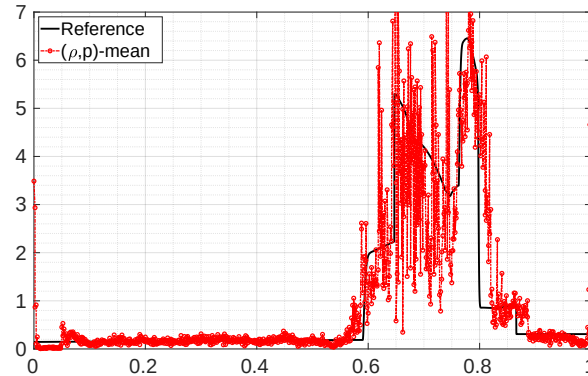
$$\nabla \cdot \mathbf{f}(u_0) = s(u_0) \quad \Rightarrow \quad u(\cdot, t) = u_0 \quad \forall t \geq 0$$

STANDARD VS. PROPERTY-PRESERVING SCHEMES

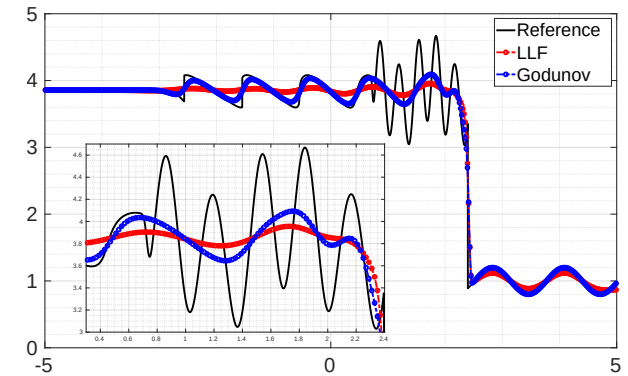
Modified Sod's shock tube: $\text{CG-}\mathbb{P}_1\text{-Roe}$



Blast wave: $\text{DG-}\mathbb{P}_1\text{-mean}$, $E = 500$

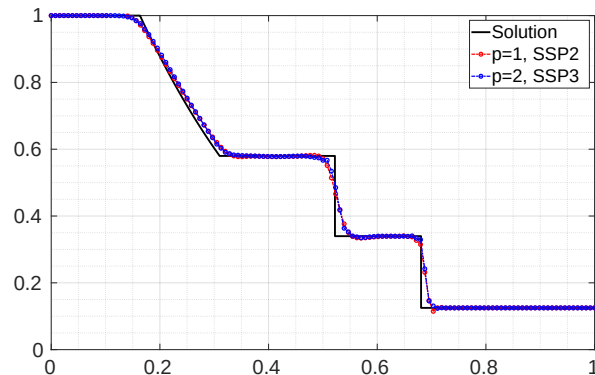


Sine-shock interaction: $\text{DG-}\mathbb{P}_0$, $E = 500$

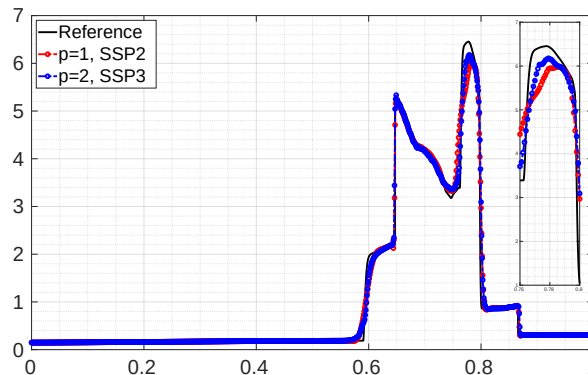


Failure of: nonentropic fluxes (left), unstable schemes (center), piecewise constant approximations (right)

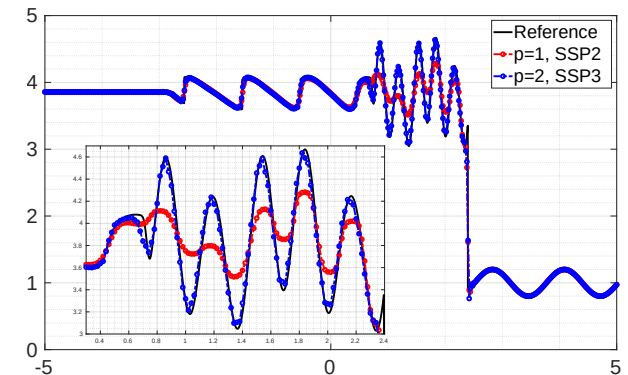
$128/p$ uniform elements



$1000/p$ uniform elements



$500/p$ uniform elements



Results obtained with physics-conforming schemes of order $p + 1$ with polynomial degrees $p \in \{1, 2\}$

Limiting for systems of conservation laws

FINITE ELEMENT DISCRETIZATION

- Mesh $\mathcal{K}_h = \{K^1, \dots, K^E\}$ over $\Omega \subset \mathbb{R}^d$, $d \in \{1, 2\}$
- Piecewise (multi-)linear continuous Galerkin space

$$V_h = \{v_h \in C(\bar{\Omega}) : v_h|_K \in \mathbb{V}_1(K) \ \forall K \in \mathcal{K}_h\}, \quad \mathbb{V}_1(K) = \begin{cases} \mathbb{P}_1(K), & K = \text{simplex}, \\ \mathbb{Q}_1(K), & K = \text{quadrilateral} \end{cases}$$

- Spatial semi-discretization

$$\int_{\Omega} w_h \left[\frac{\partial u_h}{\partial t} + \nabla \cdot \mathbf{f}(u_h) \right] d\mathbf{x} = 0 \quad \forall w_h \in V_h, \quad u_h(\mathbf{x}, t) = \sum_{j=1}^N u_j(t) \varphi_j(\mathbf{x})$$

- Group finite element approximation [Fletcher \(1983\)](#)

$$\mathbf{f}(u_h) \approx \mathbf{f}_h = \sum_{j=1}^N \mathbf{f}_j \varphi_j, \quad \mathbf{f}_j := \mathbf{f}(u_j)$$

- Matrix-vector notation

$$\sum_{j \in \mathcal{N}_i} \left[m_{ij} \frac{du_j}{dt} + \mathbf{f}_j \cdot \mathbf{c}_{ij} \right] = 0 \quad \forall i \in \{1, \dots, N\}, \quad m_{ij} = \int_{\Omega} \varphi_i \varphi_j d\mathbf{x}, \quad \mathbf{c}_{ij} = \int_{\Omega} \varphi_i \nabla \varphi_j d\mathbf{x}$$

■ Partition of unity property

$$\begin{aligned} \mathbf{c}_{ij} &= \int_{\Omega} \varphi_i \nabla \varphi_j \, d\mathbf{x}, \quad \Rightarrow \quad \mathbf{c}_{ii} = - \sum_{j \in \mathcal{N}_i \setminus \{i\}} \mathbf{c}_{ij} \\ &\Rightarrow \quad \sum_{j \in \mathcal{N}_i} m_{ij} \frac{du_j}{dt} = - \sum_{j \in \mathcal{N}_i \setminus \{i\}} (\mathbf{f}_j - \mathbf{f}_i) \cdot \mathbf{c}_{ij} \end{aligned}$$

■ Rowsum mass lumping

$$m_i \frac{du_i}{dt} = - \sum_{j \in \mathcal{N}_i \setminus \{i\}} (\mathbf{f}_j - \mathbf{f}_i) \cdot \mathbf{c}_{ij} \quad \forall i \in \{1, \dots, N\}, \quad m_i := \sum_{j \in \mathcal{N}_i} m_{ij} = \int_{\Omega} \varphi_i \, d\mathbf{x} > 0$$

■ Artificial diffusion

$$d_{ij} = \max\{\lambda_{ij} |\mathbf{c}_{ij}|, \lambda_{ji} |\mathbf{c}_{ji}|\}, \quad \lambda_{ij} = \lambda(u_i, u_j; \mathbf{n}_{ij}) \geq \max\{|\mathbf{f}'(u_i) \cdot \mathbf{n}_{ij}|, |\mathbf{f}'(u_j) \cdot \mathbf{n}_{ij}|\}, \quad \mathbf{n}_{ij} = \mathbf{c}_{ij} / |\mathbf{c}_{ij}|$$

■ Property-preserving first-order method [Abgrall \(2006\)](#), [Kuzmin et al. \(2010\)](#), [Guermond and Popov \(2016\)](#)

$$m_i \frac{du_i}{dt} = \sum_{j \in \mathcal{N}_i \setminus \{i\}} [d_{ij}(u_j - u_i) - (\mathbf{f}_j - \mathbf{f}_i) \cdot \mathbf{c}_{ij}]$$

LOCAL LAX–FRIEDRICHS (LLF) METHOD

- The resulting algebraic LLF method was analyzed by [Guermond and Popov \(2016\)](#)

$$m_i \frac{du_i}{dt} = \sum_{j \in \mathcal{N}_i \setminus \{i\}} [d_{ij}(u_j - u_i) - (\mathbf{f}_j - \mathbf{f}_i) \cdot \mathbf{c}_{ij}] = \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij}(\bar{u}_{ij} - u_i)$$

- Let $\mathcal{A}$ be the largest admissible convex set of the system

$$\mathcal{A} \text{ convex, } u_i, u_j \in \mathcal{A} \quad \Rightarrow \quad \bar{u}_{ij} := \frac{u_i + u_j}{2} + \frac{(\mathbf{f}_i - \mathbf{f}_j) \cdot \mathbf{c}_{ij}}{2d_{ij}} \in \mathcal{A}$$

- Proof: Consider projected 1D Riemann problem

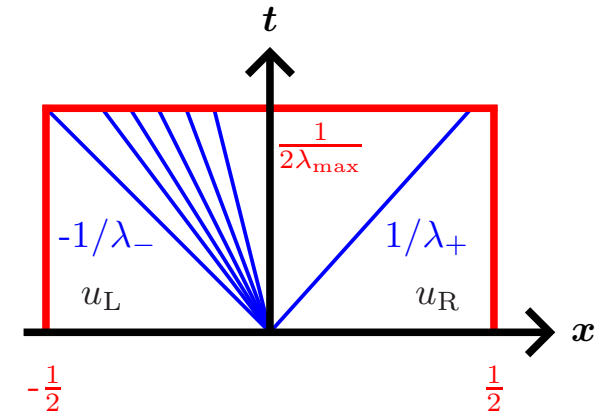
$$\frac{\partial u}{\partial t} + \frac{\partial \mathbf{f}(u) \cdot \mathbf{n}}{\partial x} = 0, \quad u_0(x) = \begin{cases} u_L & \text{if } x < 0, \\ u_R & \text{if } x > 0 \end{cases}$$

- Integrate over space-time control volume $[-\frac{1}{2}, \frac{1}{2}] \times [0, \tau]$

- Make sure that $u(\mp \frac{1}{2}, \tau) = u_{L/R}$ via $\tau \leq \frac{1}{2\lambda_{\max}}$

- Admissibility of *bar states* [Harten et al. \(1983\)](#)

$$\bar{u}_{LR} = \int_{-\frac{1}{2}}^{\frac{1}{2}} u(x, \tau) dx = \frac{u_L + u_R}{2} + \tau[\mathbf{f}(u_L) - \mathbf{f}(u_R)] \cdot \mathbf{n} \in \mathcal{A}$$



MONOLITHIC CONVEX LIMITING (MCL)

- Addition of **limited antidiffusive fluxes** Kuzmin (2020)

$$m_i \frac{du_i}{dt} = \sum_{j \in \mathcal{N}_i \setminus \{i\}} [d_{ij}(u_j - u_i) - (\mathbf{f}_j - \mathbf{f}_i) \cdot \mathbf{c}_{ij} + \mathbf{f}_{ij}^*] = \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij}(\bar{u}_{ij}^* - u_i)$$

$$\bar{u}_{ij}^* := \bar{u}_{ij} + \frac{\mathbf{f}_{ij}^*}{2d_{ij}} \stackrel{!}{\in} \mathcal{A} \quad (\text{physical constraints}) \quad \mathbf{f}_{ij}^* = -\mathbf{f}_{ji}^* \quad (\text{conservation})$$

- Discrete maximum principles in the scalar case

$$u_i^{\min} \stackrel{!}{\leq} \bar{u}_{ij} + \frac{\mathbf{f}_{ij}^*}{2d_{ij}} \stackrel{!}{\leq} u_i^{\max} \quad \Leftrightarrow \quad 2d_{ij}(u_i^{\min} - \bar{u}_{ij}) \stackrel{!}{\leq} \mathbf{f}_{ij}^* \stackrel{!}{\leq} 2d_{ij}(u_i^{\max} - \bar{u}_{ij})$$

$$u_j^{\min} \stackrel{!}{\leq} \bar{u}_{ji} - \frac{\mathbf{f}_{ij}^*}{2d_{ij}} \stackrel{!}{\leq} u_j^{\max} \quad \Leftrightarrow \quad 2d_{ij}(\bar{u}_{ji} - u_j^{\max}) \stackrel{!}{\leq} \mathbf{f}_{ij}^* \stackrel{!}{\leq} 2d_{ij}(\bar{u}_{ji} - u_j^{\min})$$

- Sequential limiting for hyperbolic systems Dobrev et al. (2018), Kuzmin (2020)

$$1. \quad h_i^{\min} \stackrel{!}{\leq} \bar{h}_{ij}^* \stackrel{!}{\leq} h_i^{\max} \quad 2. \quad \bar{h}_{ij}^* \mathbf{v}_i^{\min} \stackrel{!}{\leq} (\overline{h\mathbf{v}})_{ij}^* \stackrel{!}{\leq} \bar{h}_{ij}^* \mathbf{v}_i^{\max}$$

- Shu–Osher stage of an explicit SSP Runge–Kutta method

$$u_i^{\text{SSP}} = \underbrace{\left(1 - \frac{\Delta t}{m_i} \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij}\right)}_{\geq 0} u_i + \frac{\Delta t}{m_i} \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij} \bar{u}_{ij}^* \quad \Rightarrow \quad \Delta t \stackrel{!}{\leq} \frac{m_i}{\sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij}}$$

SEMI-DISCRETE ENTROPY FIX

- Entropy pair $(U, \mathbf{F})$, entropy variable v , entropy potential ψ

$$v(u) = U'(u), \quad \mathbf{F}'(u) = v(u)^\top \mathbf{f}'(u), \quad \psi(u) = v(u)^\top \mathbf{f}(u) - \mathbf{F}(u)$$

- Bound-preserving algebraic flux correction scheme

$$m_i \frac{du_i}{dt} = \sum_{j \in \mathcal{N}_i \setminus \{i\}} [d_{ij}(u_j - u_i) - (\mathbf{f}_j - \mathbf{f}_i) \cdot \mathbf{c}_{ij} + \textcolor{red}{f}_{ij}^*]$$

- Entropy stability condition [Tadmor \(1987\)](#), [Kuzmin and Quezada de Luna \(2020\)](#)

$$(v_j - v_i)^\top \left[(\mathbf{f}_i + \mathbf{f}_j) \cdot \frac{\mathbf{c}_{ij}}{2} + \frac{d_{ij}}{2}(u_i - u_j) - \frac{\textcolor{red}{f}_{ij}^*}{2} \right] \leq [\psi(u_j) - \psi(u_i)] \cdot \mathbf{c}_{ij}$$

implies (local and global) **semi-discrete** entropy inequalities

$$\frac{d}{dt} \int_{\Omega} \left(\sum_{i=1}^N U_i \varphi_i \right) d\mathbf{x} + \int_{\partial\Omega} \left(\sum_{i=1}^N \mathbf{F}_i \varphi_i \right) \cdot \mathbf{n} d\mathbf{s} \leq 0$$

- Enforce Tadmor's condition for any particular entropy pair $(U, \mathbf{F})$ via limiting

$$\textcolor{red}{f}_{ij}^* \rightarrow \textcolor{green}{f}_{ij}^{**} = \alpha_{ij} \textcolor{red}{f}_{ij}^* = -\textcolor{green}{f}_{ji}^{**}, \quad \alpha_{ij} = \alpha_{ji} \in [0, 1]$$

SIDE NOTE: ENERGY-STABLE NAVIER–STOKES SOLVER

- Incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} + \frac{1}{\rho} \nabla p - \nu \Delta \mathbf{v} = \mathbf{0}, \quad \nabla \cdot \mathbf{v} = 0 \quad (1)$$

- Testing (1) with $\mathbf{v}$ yields energy stability

$$\frac{d}{dt} \int_K U(\mathbf{v}) \, d\mathbf{x} \leq 0, \quad U(\mathbf{v}) = \frac{|\mathbf{v}|^2}{2}, \quad \text{for } K = \Omega \text{ (global) or } K \subset \Omega \text{ (local)}$$

- (Multi-)linear finite elements with [Becker and Hansbo \(2008\)](#)-stabilization, discretize nonlinearity via skew-symmetric form

$$\sum_{j \in \mathcal{N}_i} a_{ij}(\mathbf{v}_h) \mathbf{v}_j, \quad a_{ij}(\mathbf{v}_h) = \frac{1}{2} \int_{\Omega} \varphi_i \mathbf{v}_h \cdot \nabla \varphi_j - \varphi_j \mathbf{v}_h \cdot \nabla \varphi_i \, d\mathbf{x}$$

- We can show the following local semi-discrete energy inequality

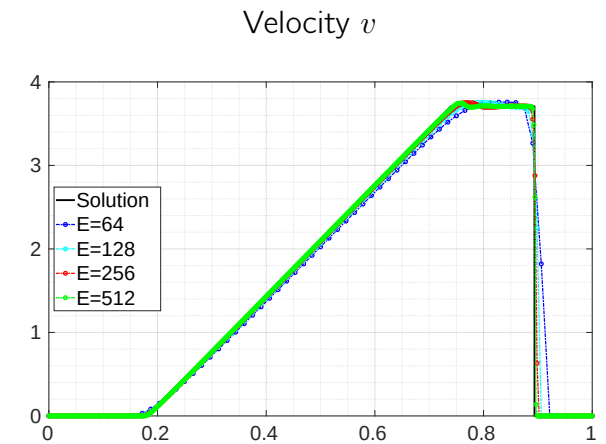
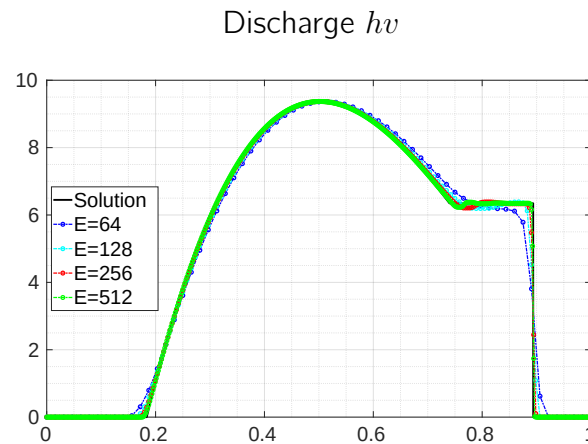
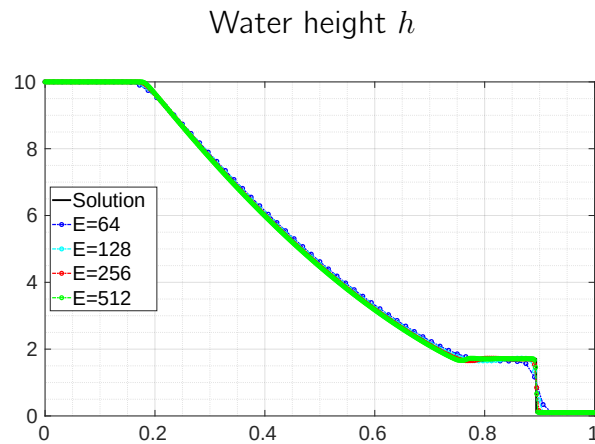
$$m_i \frac{dU(\mathbf{v}_i)}{dt} + \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2|\mathbf{c}_{ij}| F_{ij}(\mathbf{v}_i, p_i, \mathbf{v}_j, p_j; \mathbf{n}_{ij}) \leq 0, \quad F_{ij} = -F_{ji}$$

- Further details and analysis in [H., Kuzmin, Lube, Öffner, \(2025\)](#)

NEED FOR ENTROPY STABILIZATION

1D dam break with a wet bed and sonic point

$$h_0(x) = \begin{cases} 10 & \text{if } x < 0.5, \\ 0.1 & \text{if } x > 0.5, \end{cases} \quad v_0 \equiv 0$$



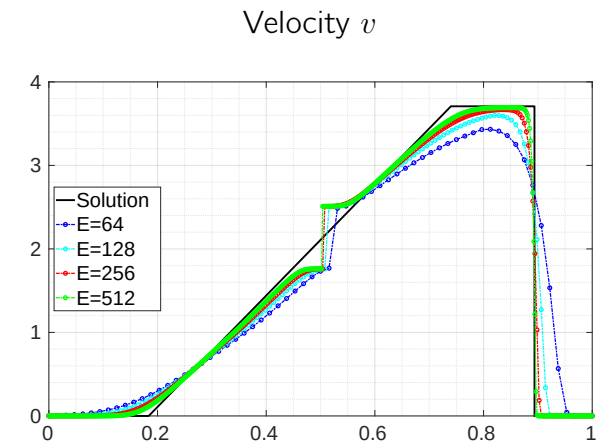
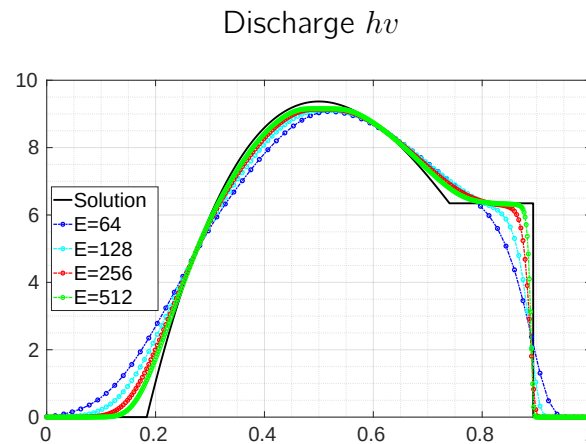
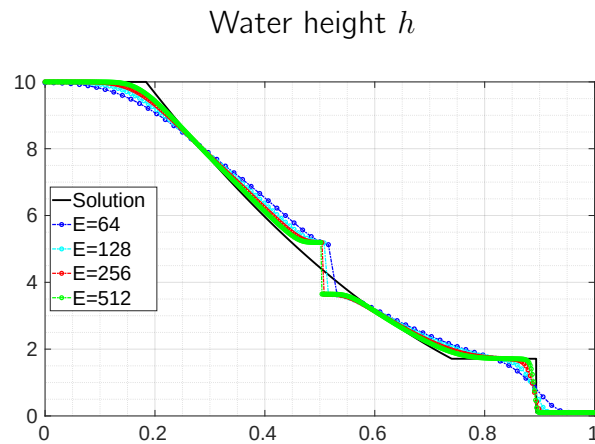
Galerkin target fluxes, bound-preserving limiting only

- No entropy shocks observed with Galerkin target fluxes
- Semi-discrete fix reduces magnitude of entropy shocks upon refinement
- Fully discrete fix prevents nonentropic results altogether [Kuzmin, H., Rupp \(2022\)](#)

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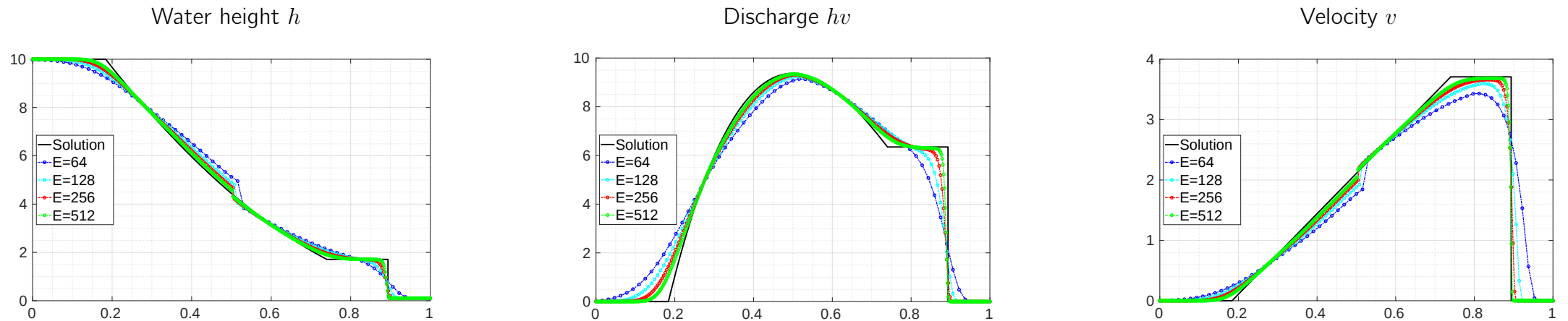
Roe target fluxes, bound-preserving limiting only

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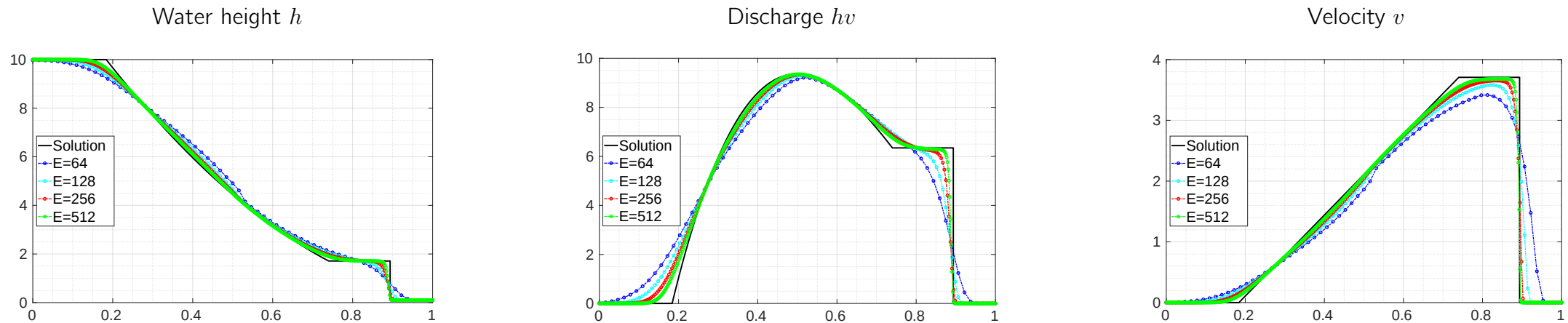
Roe target fluxes, bound-preserving limiting and semi-discrete entropy fix

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NEED FOR ENTROPY STABILIZATION

1D dam break with a wet bed and sonic point

$$h_0(x) = \begin{cases} 10 & \text{if } x < 0.5, \\ 0.1 & \text{if } x > 0.5, \end{cases} \quad v_0 \equiv 0$$



Roe target fluxes, bound-preserving limiting and fully discrete explicit entropy fix

- No entropy shocks observed with Galerkin target fluxes
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- Fully discrete fix prevents nonentropic results altogether [Kuzmin, H., Rupp \(2022\)](#)

VORTICITY ADVECTION

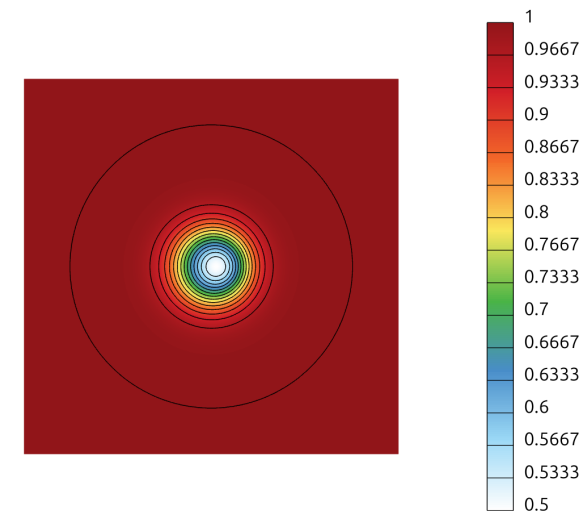
- Smooth solution for the SWE [Fjordholm \(2009\)](#)
- Periodic domain $\Omega = (-50, 50)^2$ subdivided using structured triangular meshes
- Initial condition (= solution at $T = 100$):

$$h_0(x, y) = 1 - \frac{c_1^2}{4c_2g} e^{-2c_2(x^2+y^2)}, \quad \mathbf{v}_0(x, y) = M \begin{bmatrix} \cos(\alpha) \\ \sin(\alpha) \end{bmatrix} + c_1 e^{-c_2(x^2+y^2)} \begin{bmatrix} y \\ -x \end{bmatrix},$$

$$M = \sqrt{2}, \quad c_1 = -0.1, \quad c_2 = 0.005, \quad \alpha = \frac{\pi}{4}$$

$100/h$	$L^1(\Omega)$	EOC	$L^2(\Omega)$	EOC	$L^\infty(\Omega)$	EOC
64	2.24E-02		3.19E-02		1.38E-01	
128	5.61E-03	2.00	8.72E-03	1.87	3.51E-02	1.98
256	1.35E-03	2.05	2.17E-03	2.01	1.00E-02	1.81
512	3.34E-04	2.02	5.38E-04	2.01	2.74E-03	1.87
1024	8.30E-05	2.01	1.34E-04	2.00	7.14E-04	1.94
avg.		2.02		1.97		1.90

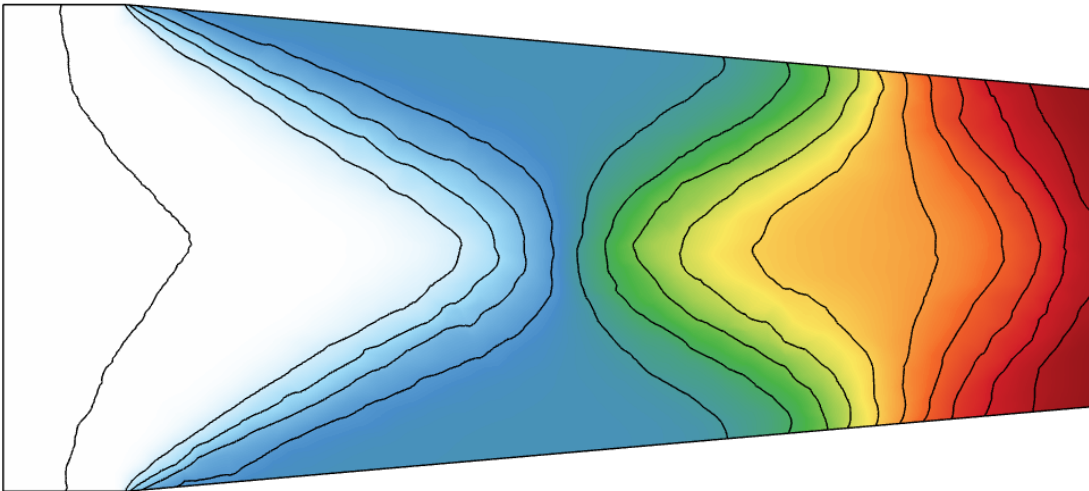
Convergence history for MCL schemes enforcing only positivity and entropy stability



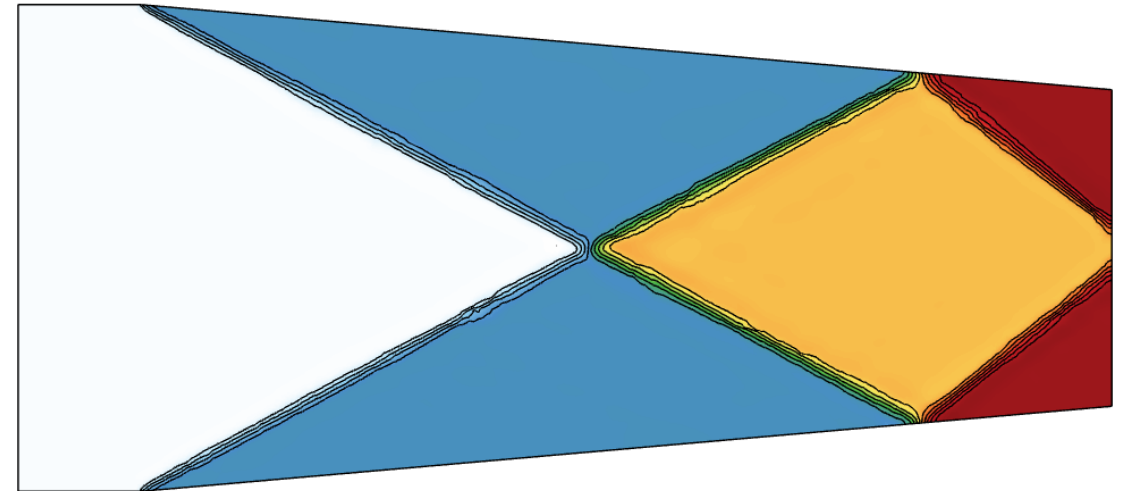
SUPERCritical CONSTRICTED CHANNEL FLOW

- Channel geometry with oblique, reflecting walls
- Shock formation and interaction, solution approaches steady state

Algebraic Local Lax–Friedrichs method (first-order)



Bound-preserving and entropy-stable MCL scheme



Approximations obtained on an unstructured triangular mesh with 50,480 cells

- Use of monolithic limiters essential for steady state convergence
- Result without entropy-limiting (not shown here) looks similar to the right panel

Extension to shallow water equations with topography

SWE WITH TOPOGRAPHY

- Hyperbolic system of balance laws

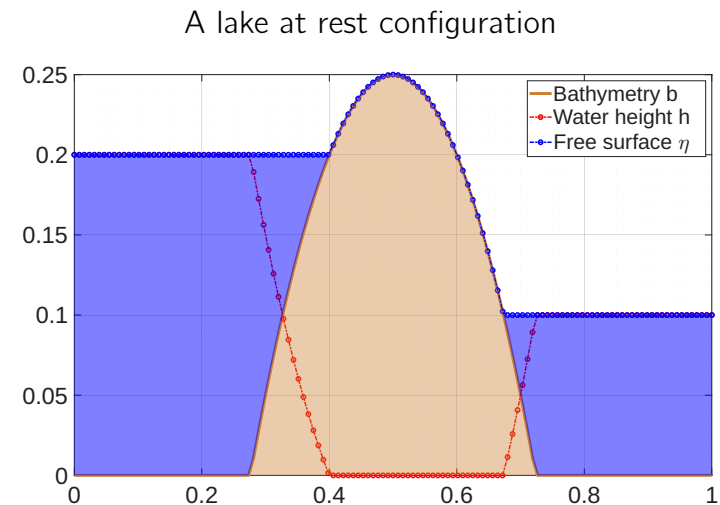
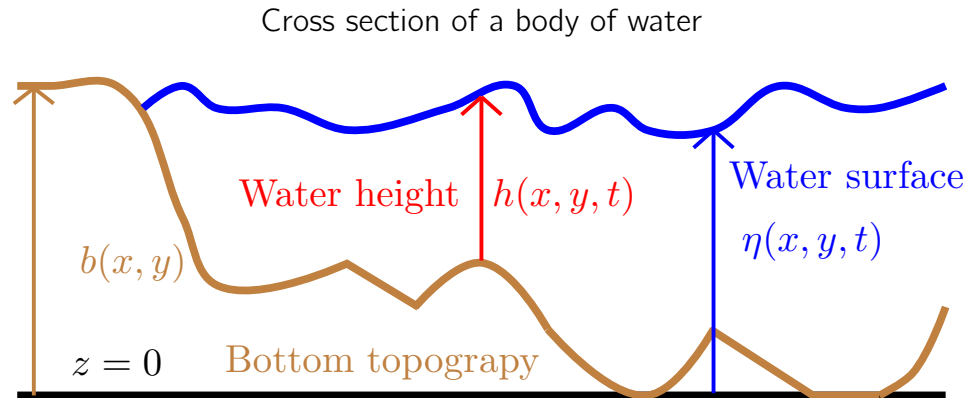
$$\frac{\partial}{\partial t} \begin{bmatrix} h \\ h\mathbf{v} \end{bmatrix} + \nabla \cdot \begin{bmatrix} h\mathbf{v}^\top \\ h\mathbf{v}\mathbf{v}^\top + \frac{g}{2}h^2\mathbf{I} \end{bmatrix} + \begin{bmatrix} 0 \\ gh\nabla b \end{bmatrix} = 0$$

- Entropy pair $(U, \mathbf{F})$ depends on bottom topography b

$$U(u, b) = \frac{1}{2} (h|\mathbf{v}|^2 + gh^2) + ghb \quad \frac{\partial U(u, b)}{\partial t} + \nabla \cdot \mathbf{F}(u, b) \leq 0$$

- Well-balancing: preserve *lake at rest*

$$\mathbf{v} \equiv \mathbf{0}, \quad \frac{g}{2}\nabla h^2 + gh\nabla b = gh\nabla\eta \equiv \mathbf{0}$$



LOW ORDER METHOD FOR SWE

- Well-balanced discretization of source term

$$\int_{\Omega} \varphi_i g h \nabla b \, d\mathbf{x} \approx \sum_{j \in \mathcal{N}_i \setminus \{i\}} g \frac{h_i + h_j}{2} (b_j - b_i) \mathbf{c}_{ij}$$

- Modified continuity equation

$$m_i \frac{dh_i}{dt} = \sum_{j \in \mathcal{N}_i \setminus \{i\}} [d_{ij}(h_j - h_i + \alpha_{ij}^b (b_j - b_i)) - (\mathbf{f}_j^h - \mathbf{f}_i^h) \cdot \mathbf{c}_{ij}]$$

- Required for well-balancing and positivity preservation Audusse et al. (2015)

$$\bar{h}_{ij}^b := \bar{h}_{ij} + \frac{\alpha_{ij}^b}{2} (b_j - b_i) \stackrel{!}{\geq} 0$$

- Additional term in momentum equation required for entropy stability

$$\begin{aligned} m_i \frac{d(h\mathbf{v})_i}{dt} = & \sum_{j \in \mathcal{N}_i \setminus \{i\}} [d_{ij}((h\mathbf{v})_j - (h\mathbf{v})_i + \frac{\mathbf{v}_i + \mathbf{v}_j}{2} \alpha_{ij}^b (b_j - b_i)) \\ & - (\mathbf{f}_j^{hv} - \mathbf{f}_i^{hv}) \cdot \mathbf{c}_{ij} - g \frac{h_i + h_j}{2} \alpha_{ij}^b (b_j - b_i) \mathbf{c}_{ij}] \end{aligned}$$

PROPERTY-PRESERVING MCL FOR SWE

- Fully discrete MCL scheme

$$u_i^{\text{SSP}} = \left(1 - \frac{\Delta t}{m_i} \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij}\right) u_i + \frac{\Delta t}{m_i} \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij} \bar{u}_{ij}^{b,*}$$

- Bar states

$$\bar{h}_{ij}^{b,*} = \bar{h}_{ij} + \frac{\alpha_{ij}^b (b_j - b_i)}{2} + \frac{\beta_{ij} f_{ij}^{h,*}}{2d_{ij}}$$

$$\overline{(h\mathbf{v})}_{ij}^{b,*} = \overline{(h\mathbf{v})}_{ij} + \frac{\mathbf{v}_i + \mathbf{v}_j}{4} \alpha_{ij}^b (b_j - b_i) - \frac{g \frac{h_i + h_j}{2} \alpha_{ij}^b (b_j - b_i) \mathbf{c}_{ij}}{2d_{ij}} + \frac{\beta_{ij} \mathbf{f}_{ij}^{h\mathbf{v},*}}{2d_{ij}}$$

- Generalized entropy limiter β_{ij} enforces local entropy inequalities, which imply

$$\frac{d}{dt} \int_{\Omega} \left(\sum_{i=1}^N U_i \varphi_i \right) d\mathbf{x} + \int_{\partial\Omega} \left(\sum_{i=1}^N \mathbf{F}_i \varphi_i \right) \cdot \mathbf{n} d\mathbf{s} \leq 0$$

- Provably well-balanced, positivity-preserving, entropy-stable scheme
- Reduces to MCL for systems of conservation laws if $b \equiv 0$

NEW WETTING AND DRYING MODELS

- First new approach: Entropy/velocity limiting

$$U(u, 0) = \frac{1}{2} (gh^2 + h|\mathbf{v}|^2) \leq U^{\max} \quad \Rightarrow \quad \text{bound for } |\mathbf{v}|$$

- ▶ Local bounds for total energy

$$U(u_i^{\text{SSP}}) \leq \left(1 - \frac{\Delta t}{m_i} \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij}\right) U(u_i) + \frac{\Delta t}{m_i} \sum_{j \in \mathcal{N}_i \setminus \{i\}} 2d_{ij} U(\bar{u}_{ij}^{b,*}) =: U_i^{\max}$$

- ▶ Convexity-based proof carries over to SSP time stepping schemes

- Second new approach: Boundary layer approximation

$$gh\nabla\eta + c\mathbf{v} \approx 0, \quad \mathbf{v}^{\text{BL}} = -\frac{g}{c}h\nabla\eta$$

- ▶ ‘Wall functions’ using the ‘friction velocity’ $\mathbf{v}^{\text{BL}}$

$$\tilde{\mathbf{v}}_i = \frac{(h\mathbf{v})_i}{\max\{h_i, \delta\}} + \max\left\{0, \frac{\delta - h_i}{\delta}\right\} \mathbf{v}_i^{\text{BL}}, \quad \delta = 10^{-3}$$

- ▶ *When the numerics fails, substitute the physics* [Zalesak \(2012\)](#)

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STEADY STATE PROBLEMS

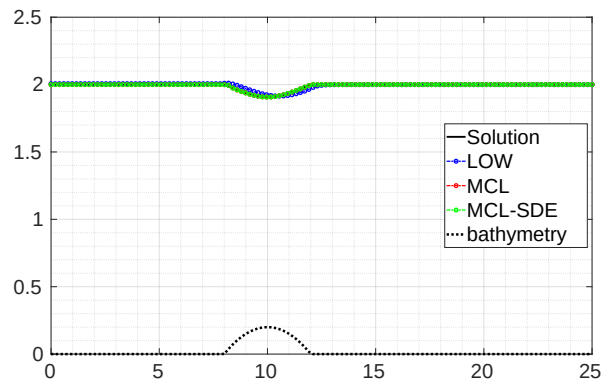
■ Acronyms

- ▶ low order method (LOW)
- ▶ monolithic convex limiting (MCL)
- ▶ monolithic convex limiting with semi-discrete entropy fix (MCL-SDE)

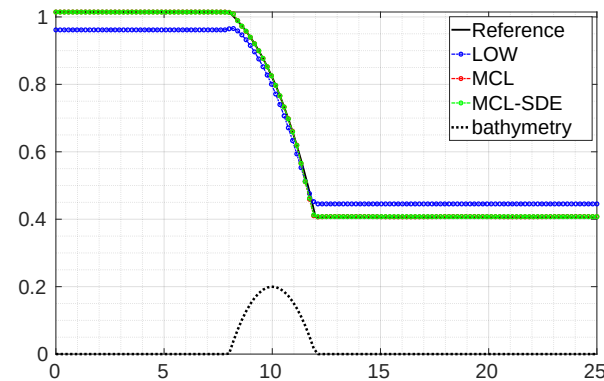
■ Moving water equilibria Vázquez-Cendón (1999)

$$hv \equiv \text{const}, \quad \frac{v^2}{2} + g(h + b) \equiv \text{const}$$

Subcritical flow

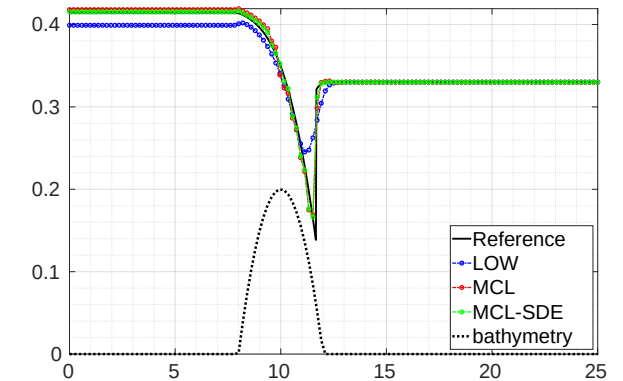


Transcritical flow w/o shock

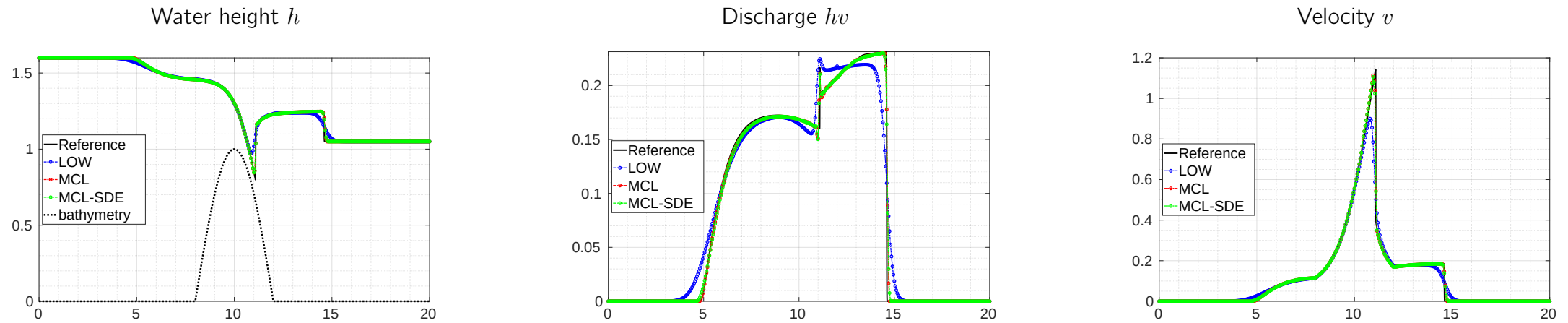


Water heights h , 128 elements

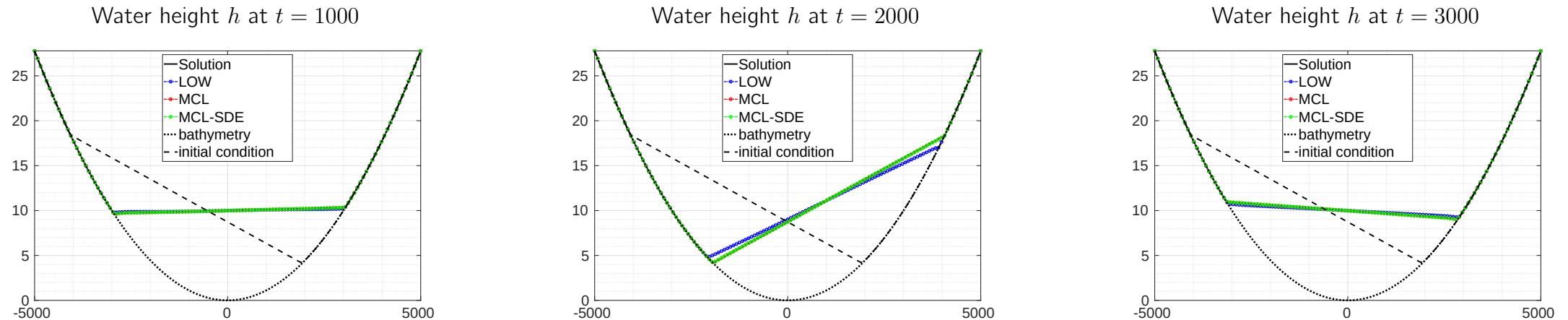
Transcritical flow with shock



TIME-DEPENDENT PROBLEMS



Dam break over topography [Winters and Gassner \(2015\)](#), wetting and drying not activated, 400 elements



Parabolic basin [Thacker \(1981\)](#), friction-based wetting and drying, 128 elements

CONCLUSIONS

Summary

- Property-preserving schemes using continuous and discontinuous finite elements
- Well-balancedness with respect to lake at rest scenarios
- Positivity preservation for the water height provably satisfied under a CFL condition
- Local entropy stability enforceable

Future prospects

- Use as forward solver in an inverse bathymetry reconstruction problem [Ruppenthal and Kuzmin \(2026\)](#)
- DG methods for systems with source terms
- Extension to multi-layer SWE
- Incorporation of techniques into realistic models

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